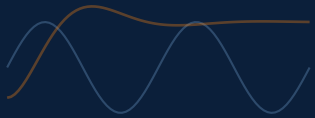
DAY 3 / 5

Mechatronics: Electrical Machines, Control, Servo Systems and Sensing



Advanced Control and Introduction to Servo Systems

Filters & compensation · state space · adaptive/robust · discrete-time control · servo architecture · inverter & DC bus

Mechatronics Training Programme

Theoretical Course Series ▪ Day 3 of 5

Agenda and learning objectives

Code	Session	Time
2.6	Filters and compensation in control loops	60 min
2.7	Introduction to modern (state-space) control	60 min
2.8	Adaptive and robust control (concepts)	60 min
2.9	Discrete-time control	60 min
3.1	Servo system architecture	30 min
3.2a	Power electronics I: rectifier, DC bus, inverter	30 min
	Net teaching time	**5 h**

Short breaks (≈10 min) after each 1-h block are not included in the net time.

After today you can . . .

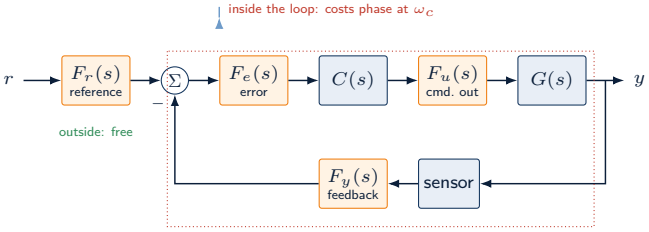
- quantify the phase cost of LPF/notch filters and design lead/lag compensators
- place poles with state feedback and build a Luenberger observer
- explain MRAC, Lyapunov adaptation, gain scheduling and the robust-stability test $|WT| < 1$
- discretise controllers correctly and budget sampling delay
- sketch a servo drive from mains to motor shaft

Notation (whole course): plant $G(s)$, controller $C(s)$, loop $L(s) = C(s)G(s)$, crossover ω_c, phase margin PM, gain margin GM, $S = 1/(1+L)$, $T = L/(1+L)$. Mechanical: J [kg m^2], B [N m s/rad], torque T_m, load torque T_L.

Session 1

2.6 Filters and Compensation in Control Systems

Why filters in a servo loop — and where they sit



What we filter against

- encoder quantisation & current-sensor noise
- mechanical resonances (couplings, belts, frames)
- DC offsets of analogue sensors
- steps in the reference (jerk, overshoot)

Placement rules

- F_r: shapes response only — **no stability cost**
- F_e, F_u, F_y: **in the loop**, their phase adds to $\angle L(j\omega_c)$
- F_y also hides the true y from the controller

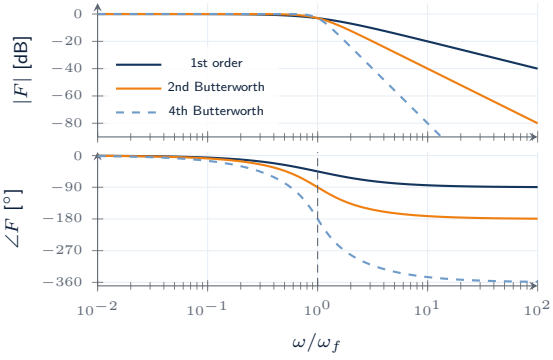
Rule

Every in-loop filter is part of $L(s)$: design it together with $C(s)$, never "add it later".

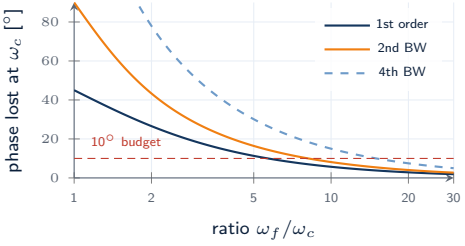
Low-pass filter: cut-off, order, roll-off, phase

$$F_1(s) = \frac{\omega_f}{s + \omega_f}, \qquad F_2(s) = \frac{\omega_f^2}{s^2 + 2\zeta\omega_f s + \omega_f^2}$$

- **cut-off** ω_f: -3 dB point ($\zeta = 1/\sqrt{2}$)
- **order** n: roll-off $-20n$ dB/dec
- **phase**: $-45°n$ at ω_f (Butterworth), $\to -90°n$
- **Butterworth**: maximally flat, $\zeta = 0.707$
- $\zeta < 0.7$: peaking — a filter can **amplify** near ω_f
- group delay at DC: $\tau_g = 1/\omega_f$ (1st), $\sqrt{2}/\omega_f$ (2nd BW)



The price of filtering: phase at crossover and the noise–bandwidth trade-off



$$\Delta\varphi_1 = \arctan\frac{\omega_c}{\omega_f} \;\approx\; \omega_c\,\tau_g \quad (\omega_f \gg \omega_c)$$

Worked example. Velocity loop $f_c = 150$ Hz, allowed loss $10°$: $f_f \geq f_c/\tan 10° = 5.67\,f_c \approx 850$ Hz (1st order). A 2nd-order BW filter needs $f_f \gtrsim 1.2$ kHz.

Noise path: meas. noise n reaches the torque command via

$$u = -\,C(s)S(s)\,n \;\approx\; -C(s)\,n \quad (\omega \gg \omega_c)$$

- high $K_p/K_d \Rightarrow$ high-frequency noise in torque, audible, heating
- lower $\omega_f \Rightarrow$ quieter, but less PM $\Rightarrow$ must lower ω_c

Trade-off

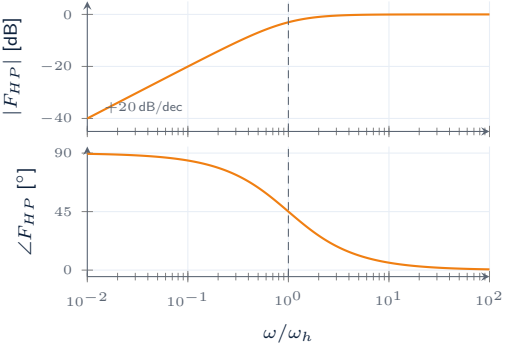
Bandwidth, noise and filter phase form a triangle: you can pick two. Typical: $\omega_f \approx (5\ldots10)\,\omega_c$.

High-pass filter: DC and offset rejection

$$F_{HP}(s) = \frac{s}{s+\omega_h} \qquad \angle F_{HP}(j\omega) = 90^\circ - \arctan\frac{\omega}{\omega_h}$$

- blocks DC: removes **offsets** and slow drift (analogue current/Hall sensors, accelerometers, strain gauges)
- **washout**: passes only changes — e.g. acceleration or vibration feedback that must not fight a constant load
- step response decays with $\tau = 1/\omega_h$
- **phase lead** below ω_h (up to $+90^\circ$)

Pitfall: an HPF in the *main* feedback path destroys DC information — the loop can no longer hold position against a constant load, and integral action becomes useless.

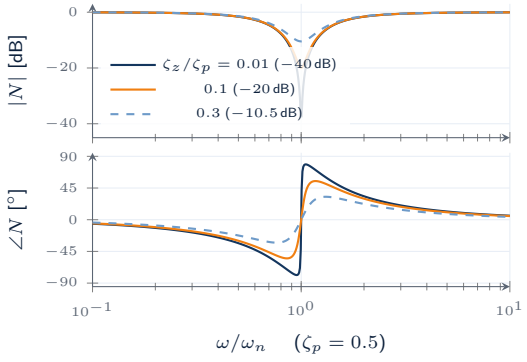


Notch filter: centre, width, Q and depth

$$N(s) = \frac{s^2 + 2\zeta_z\omega_n s + \omega_n^2}{s^2 + 2\zeta_p\omega_n s + \omega_n^2}, \qquad \zeta_z < \zeta_p$$

centre	ω_n (set to the resonance)
depth	$\lvert N(j\omega_n)\rvert = \zeta_z/\zeta_p$
width	$\Delta\omega_{-3\,\mathrm{dB}} \approx 2\zeta_p\omega_n$
quality	$Q = 1/(2\zeta_p) = \omega_n/\Delta\omega$

- unity gain far away; **phase lag below** ω_n, lead above
- wide notch $\Rightarrow$ robust to resonance drift, but more phase loss at ω_c
- narrow, deep notch $\Rightarrow$ cheap in phase, but fails if ω_r moves



Notch in practice: suppressing a 1 kHz mechanical resonance

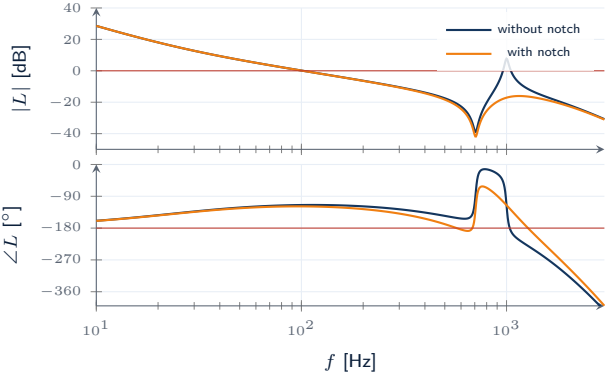
Two-mass axis (motor J_M – shaft – load J_L), velocity measured on the motor:

$$G(s) = \frac{1}{(J_M + J_L)s} \cdot \frac{s^2/\omega_a^2 + 2\zeta s/\omega_a + 1}{s^2/\omega_r^2 + 2\zeta s/\omega_r + 1}$$

$J_M = J_L = 1 \times 10^{-3}\,\text{kg m}^2$, $f_r = 1\,\text{kHz}$, $f_a = f_r/\sqrt{2}$, $\zeta = 0.02$.
PI with $f_c = 100\,\text{Hz}$; delay $1.5T$ at 8 kHz; 2 kHz current loop.

	no notch	with notch
crossovers	3	1
PM at 100 Hz	65°	61°
PM at ≈1.04 kHz	-4°	—
closed loop	unstable	stable

Notch: $\omega_n = \omega_r$, $\zeta_z = 0.02$, $\zeta_p = 0.35$ ($-25\,\text{dB}$); costs only $\approx 4^\circ$ at f_c.



Model and numbers computed for this course; cf. [Ellis, 2012, ch. 16], [Franklin et al., 2019].

Phase-lead compensator

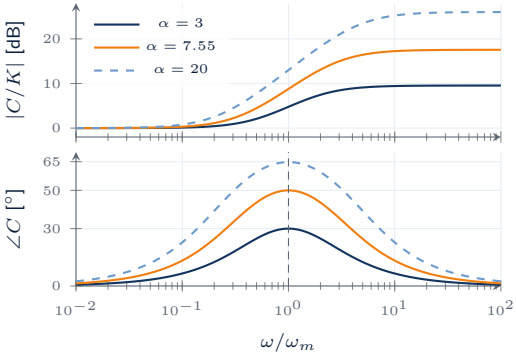
$$C_{\text{lead}}(s) = K\,\frac{1+\alpha T s}{1+Ts}, \qquad \alpha > 1$$

Zero at $1/(\alpha T)$, pole at $1/T$. Maximum phase at the geometric mean:

$$\omega_m = \frac{1}{T\sqrt{\alpha}}, \quad \sin\varphi_{\max} = \frac{\alpha-1}{\alpha+1}, \quad \alpha = \frac{1+\sin\varphi_{\max}}{1-\sin\varphi_{\max}}$$

- gain at ω_m: $K\sqrt{\alpha}$ $(10\log_{10}\alpha\,\text{dB})$; at HF: $K\alpha$
- $\alpha = 3 \to 30°$, $7.5 \to 50°$, $20 \to 65°$
- $\varphi_{\max} > 60°$: HF gain explodes — use two stages
- a lead is a **band-limited PD**: $K_d s/(Ts+1)$ plus K_p

[Ogata, 2010, ch. 7]; [Franklin, Powell & Emami-Naeini, 2019, ch. 6].

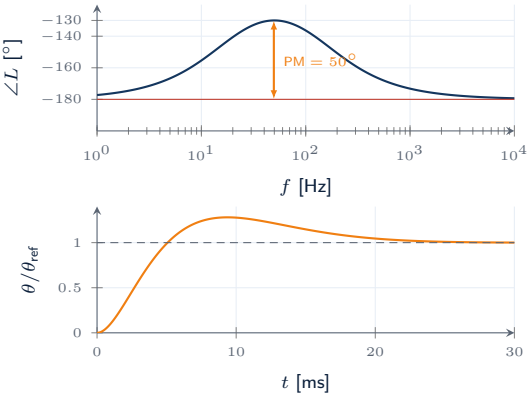


Lead design — worked example: position loop on a pure inertia

Plant (ideal torque loop): $G(s) = \dfrac{1}{Js^2}$, $J = 2 \times 10^{-4}$ kg m^2.
A pure gain gives $\angle L = -180^\circ \Rightarrow$ PM $= 0$.
Spec: $f_c = 50$ Hz ($\omega_c = 314.2$ rad/s), PM $= 50^\circ$.

1. $\alpha = \dfrac{1 + \sin 50^\circ}{1 - \sin 50^\circ} = 7.55$
2. put $\omega_m = \omega_c$: $T = \dfrac{1}{\omega_c \sqrt{\alpha}} = 1.159$ ms
   zero $1/(\alpha T) = 114$ rad/s (18.2 Hz), pole $1/T = 863$ rad/s (137 Hz)
3. $|CG| = 1$ at ω_c: $K = \dfrac{J\omega_c^2}{\sqrt{\alpha}} = 7.18$ N m/rad

Check (computed): $f_c = 49.5$ Hz, PM $= 50.0^\circ$, GM $= \infty$.
Step overshoot **28 %** — higher than the 2nd-order estimate (≈17 %) because of the lead zero; a reference prefilter F_r removes it.



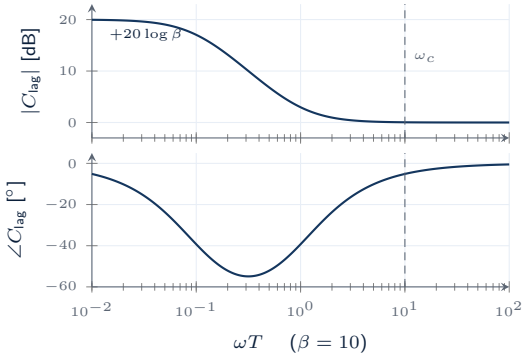
Phase-lag compensator and lead–lag

$$C_{\text{lag}}(s) = \beta\,\frac{1+Ts}{1+\beta Ts}, \qquad \beta > 1$$

- low-frequency gain $\times\beta \Rightarrow$ steady-state error $\div\beta$, **stiffer** against load torque
- HF gain unchanged $\Rightarrow \omega_c$ and noise unchanged
- max. lag $\sin\varphi = (\beta-1)/(\beta+1)$ at $1/(T\sqrt{\beta})$
- rule: zero $1/T \approx \omega_c/10 \Rightarrow$ phase cost at ω_c $= \arctan 0.1 - \arctan(0.1/\beta) = 5.1^\circ$ for $\beta = 10$

Lead–lag: $C = C_{\text{lead}}C_{\text{lag}}$ — lead buys PM at ω_c, lag buys LF gain.

PID $\equiv$ lead–lag: PI = lag with pole at $s = 0$ ($\beta \to \infty$); filtered D = lead.



Session 2.6 summary: filter and compensator scope list

Element	Design parameters	Typical servo use	Cost / risk
LPF	cut-off ω_f, order n, ζ; roll-off $-20n$ dB/dec	encoder/current noise, torque-command smoothing	phase $\approx \omega_c/\omega_f$ rad per order
HPF	corner ω_h	offset removal, washout, AC-coupled sensors	kills DC info if in main loop
Notch	centre ω_n, width $2\zeta_p\omega_n$, Q, depth ζ_z/ζ_p	mechanical resonance, belt/coupling modes	phase lag below ω_n; detuning
Lead	α, T ($\varphi_{\max}$, ω_m)	add PM for inertia/double-integrator plants	HF gain $\times\alpha \Rightarrow$ noise
Lag	β, T	stiffness, steady-state accuracy	slow tail; small PM loss

Key take-aways

- Filters in the loop are part of $L(s)$: budget their phase at ω_c (rule: $\omega_f \geq 5$–$10\,\omega_c$).
- Notch filters convert an unstable resonant loop into a well-damped one at a few degrees of PM.
- Lead: α from $\varphi_{\max}$, centre at ω_c; lag: zero a decade below ω_c.

Session 2

2.7 Introduction to Modern Control

State-space model of a servo axis

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + Bu, \qquad y = C\boldsymbol{x} + Du$$
$$G(s) = C\,(sI - A)^{-1}B + D$$

Axis with ideal torque loop, $J\dot{\omega} = -B_v\omega + u - T_L$, $\boldsymbol{x} = [\theta \;\; \omega]^\mathsf{T}$, $u = T_m$:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -a \end{bmatrix},\; B = \begin{bmatrix} 0 \\ b \end{bmatrix},\; C = \begin{bmatrix} 1 & 0 \end{bmatrix},\quad a = \tfrac{B_v}{J},\; b = \tfrac{1}{J}$$

$$\Rightarrow G(s) = \frac{1}{s(Js + B_v)}.$$

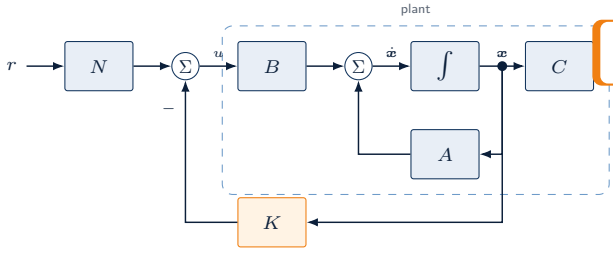
Numbers used in this session: $J = 1 \times 10^{-3}\,\mathrm{kg\,m^2}$, $B_v = 2 \times 10^{-3}\,\mathrm{N\,m\,s/rad} \Rightarrow a = 2\,\mathrm{s^{-1}}$, $b = 1000$.

Including the electrical state (DC/PM motor, $\boldsymbol{x} = [\theta \;\; \omega \;\; i]^\mathsf{T}$, $u = v$):

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{B_v}{J} & \frac{k_t}{J} \\ 0 & -\frac{k_e}{L} & -\frac{R}{L} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L} \end{bmatrix}$$

- state = **memory** of the system (energy stores: L, J, position)
- works for MIMO, time-domain design, observers
- transfer function hides internal (e.g. cancelled) modes — the state model does not

State feedback and pole placement



$$u = -K\boldsymbol{x} + Nr \;\Rightarrow\; \dot{\boldsymbol{x}} = (A - BK)\boldsymbol{x} + BN\,r$$

- closed-loop poles $= \mathrm{eig}(A - BK)$
- if (A, B) is **controllable**, **all** poles can be placed arbitrarily
- N scales the reference for zero steady-state error (nominal)
- all states must be measured — or **estimated** (observer)

Choosing poles: from ω_n, ζ of a dominant pair (overshoot, settling), extra poles 3–5$\times$ faster.

Limits: actuator saturation, sensor noise, sampling rate and unmodelled resonances bound how fast poles can go.

Controllability and Ackermann's formula

Controllability

(A, B) is controllable $\iff$

$$\operatorname{rank} \mathcal{C} = n, \qquad \mathcal{C} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}$$

- every state can be driven anywhere in finite time
- an uncontrollable mode keeps its eigenvalue whatever K is
- servo example: a load mode whose *node* sits at the motor cannot be damped from the motor torque

[Ogata, 2010, ch. 10]; [Åström & Murray, 2021, ch. 7].

Ackermann (single input)

$$K = \begin{bmatrix} 0 & \cdots & 0 & 1 \end{bmatrix} \mathcal{C}^{-1} \varphi_d(A)$$

$\varphi_d(s) = \prod_i (s - p_i)$: desired characteristic polynomial, evaluated with the matrix A.

Servo axis:

$$\mathcal{C} = \begin{bmatrix} 0 & b \\ b & -ab \end{bmatrix} = \begin{bmatrix} 0 & 1000 \\ 1000 & -2000 \end{bmatrix}$$

$\det \mathcal{C} = -b^2 = -10^6 \neq 0 \Rightarrow$ controllable.

Worked example: placing the poles of the servo axis

Desired: $\omega_n = 100$ rad/s, $\zeta = 0.7 \Rightarrow \varphi_d(s) = s^2 + 140s + 10^4$, poles $-70 \pm j71.4$.

$$A^2 = \begin{bmatrix} 0 & -2 \\ 0 & 4 \end{bmatrix}, \quad \varphi_d(A) = A^2 + 140A + 10^4 I = \begin{bmatrix} 10^4 & 138 \\ 0 & 9724 \end{bmatrix}$$

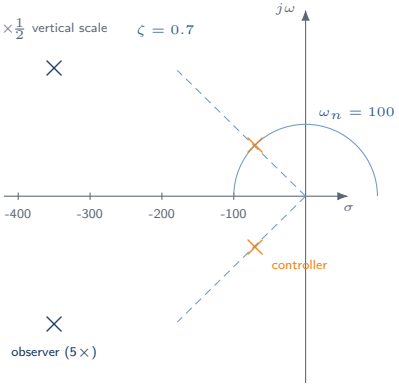
$$\mathcal{C}^{-1} = \begin{bmatrix} 0.002 & 0.001 \\ 0.001 & 0 \end{bmatrix}, \quad [0\ 1]\,\mathcal{C}^{-1} = [0.001 \quad 0]$$

$$K = [0.001 \quad 0]\,\varphi_d(A) = [\,10 \quad 0.138\,]$$

$k_1 = 10$ N m/rad (stiffness), $k_2 = 0.138$ N m s/rad (damping).

Check:
$\det(sI - A + BK) = s^2 + (a + bk_2)s + bk_1 = s^2 + 140s + 10^4$ ✓

$\Rightarrow$ for this plant state feedback **is** a PD position controller with velocity feedback; $N = k_1$ gives $\theta \to r$.



Observability and the Luenberger observer

Observability

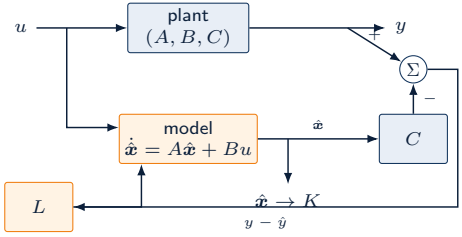
$$\operatorname{rank} \mathcal{O} = n, \quad \mathcal{O} = \begin{bmatrix} C;\ CA;\ \cdots;\ CA^{n-1} \end{bmatrix}$$

- $y = \theta$ (encoder): $\mathcal{O} = I \Rightarrow$ observable
- $y = \omega$ only: $\mathcal{O} = \left[\begin{smallmatrix} 0 & 1 \\ 0 & -2 \end{smallmatrix}\right]$, rank 1 $\Rightarrow$ θ **unobservable**

$$\dot{\hat{\boldsymbol{x}}} = A\hat{\boldsymbol{x}} + Bu + L\,(y - C\hat{\boldsymbol{x}})$$
$$\boldsymbol{e} = \boldsymbol{x} - \hat{\boldsymbol{x}}: \quad \dot{\boldsymbol{e}} = (A - LC)\,\boldsymbol{e}$$

$\operatorname{eig}(A - LC)$: 2–6× faster than controller poles (dual Ackermann on $(A^{\mathsf{T}}, C^{\mathsf{T}})$).

[Luenberger, 1964]; [Franklin, Powell & Emami-Naeini, 2019, ch. 7].



The observer is a *simulation* of the plant, continuously corrected by the measured output.

Observer example: velocity from a 16-bit encoder

Same axis, $y = \theta$ quantised to $2\pi/2^{16}$, sampled at 4 kHz.

Observer poles: $\omega_o = 500\,\text{rad/s}$, $\zeta = 0.7$ (5× controller):

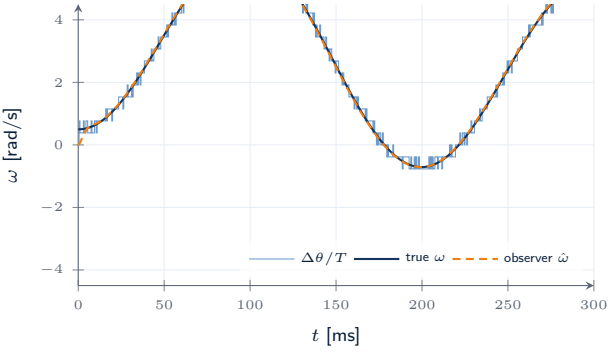
$$\det(sI - A + LC) = s^2 + (a + l_1)s + (a\,l_1 + l_2)$$

$$\stackrel{!}{=} s^2 + 700s + 2.5 \cdot 10^5$$

$\Rightarrow l_1 = 698\,\text{s}^{-1}$, $l_2 = 2.486 \times 10^5\,\text{s}^{-2}$.

Finite difference $\Delta\theta/T$: one count $= 0.38\,\text{rad/s} \Rightarrow$ staircase noise.

Observer: smooth $\hat{\omega}$, initial error ($\hat{\omega}(0) = 0$ vs. $0.5\,\text{rad/s}$) dies in $\approx 10\,\text{ms}$, and uses the torque command $u \Rightarrow$ **less phase lag** than differentiation + LPF.



Simulation for this course; observer-based velocity estimation: [Ellis, 2012, ch. 18].

The separation principle

Feed back the estimate:
$u = -K\hat{\boldsymbol{x}} + Nr = -K\boldsymbol{x} + K\boldsymbol{e} + Nr.$

$$\begin{bmatrix} \dot{\boldsymbol{x}} \\ \dot{\boldsymbol{e}} \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{e} \end{bmatrix} + \begin{bmatrix} BN \\ 0 \end{bmatrix} r$$

Block-triangular $\Rightarrow$

$$\text{eig}(\text{closed loop}) = \text{eig}(A - BK) \ \cup \ \text{eig}(A - LC)$$

Design K and L independently.

Observer + state feedback together form an ordinary dynamic compensator $C(s)$ of order n.

Caveats in real servos

- exact only for a perfect model: errors in J, B_v, k_t bias $\hat{\boldsymbol{x}}$
- no guaranteed stability margins for observer-based (LQG) designs [Doyle, 1978] — always check PM/GM of the resulting $L(s)$
- load torque is unmodelled $\Rightarrow$ add it as an extra state $\dot{T}_L = 0$ (disturbance-estimating observer)
- observer runs in the sampled controller: discretise it properly (Session 2.9)

Integral action in state feedback

Augment with the integral of the tracking error:

$$\dot{z} = r - y, \qquad u = -k_1\theta - k_2\omega + k_I z$$

$$\det = s^3 + (a + bk_2)s^2 + bk_1 s + bk_I$$

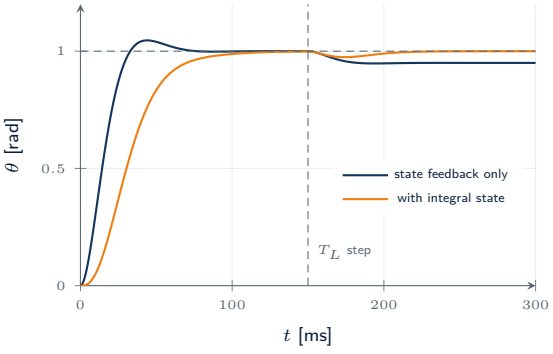
Desired
$(s+50)(s^2+140s+10^4) = s^3+190s^2+17000s+5\cdot10^5$:

$$k_1 = 17, \quad k_2 = 0.188, \quad k_I = 500\ \text{N m}/(\text{rad s})$$

Test: 1 rad step at $t = 0$, load step $T_L = 0.5$ N m at 150 ms.

- without z: error $= T_L/k_1 = 0.05$ rad (2.9°) remains
- with z: error $\to 0$ — **type-1 robustness** to constant loads and parameter errors

Implement with anti-windup (Session 2.9).



Optimal state feedback: LQR in one slide

Instead of choosing poles, choose a **cost**:

$$J = \int_0^\infty \left(\boldsymbol{x}^\mathsf{T} Q \boldsymbol{x} + u^\mathsf{T} R u \right) dt, \quad Q \succeq 0,\ R \succ 0$$

Optimal law $u = -K\boldsymbol{x}$ with

$$K = R^{-1} B^\mathsf{T} P, \qquad A^\mathsf{T} P + PA - PBR^{-1}B^\mathsf{T}P + Q = 0$$

(continuous algebraic Riccati equation; solved numerically).

[Anderson & Moore, 1990]; [Åström & Murray, 2021, ch. 7]; [Doyle, 1978].

- **Bryson's rule**: $Q_{ii} = 1/x_{i,\max}^2$, $R = 1/u_{\max}^2$ — weights in physical units
- increasing $Q/R \Rightarrow$ faster, more aggressive
- SISO, full-state LQR guarantees PM $\geq 60^\circ$ and GM $\in [\frac{1}{2}, \infty)$
- the observer counterpart is the **Kalman filter**; together: LQG (margins **not** guaranteed)

In drives LQR is used for multi-mass axes and gantry/crane anti-sway, where poles are hard to choose by hand.

Session 2.7 summary

Key take-aways

- State model $\dot{\boldsymbol{x}} = A\boldsymbol{x} + Bu$ exposes every energy store and mode.
- Controllable $\Rightarrow$ all poles placeable by $u = -K\boldsymbol{x}$ (Ackermann).
- Observable $\Rightarrow$ unmeasured states estimated by $\dot{\hat{\boldsymbol{x}}} = A\hat{\boldsymbol{x}} + Bu + L(y - C\hat{\boldsymbol{x}})$.
- Separation: K and L designed independently — but verify margins.
- Integral state removes steady error under load.
- LQR: choose weights, get gains with good margins.

Tool	Test / formula
controllability	rank $[B\ AB\cdots]$
observability	rank $[C;\ CA;\cdots]$
pole placement	$K = [0\cdots1]\mathcal{C}^{-1}\varphi_d(A)$
observer	$\mathrm{eig}(A - LC)$ 2–6$\times$ faster
integral	augment $\dot{z} = r - y$
LQR	$K = R^{-1}B^{\mathsf{T}}P$

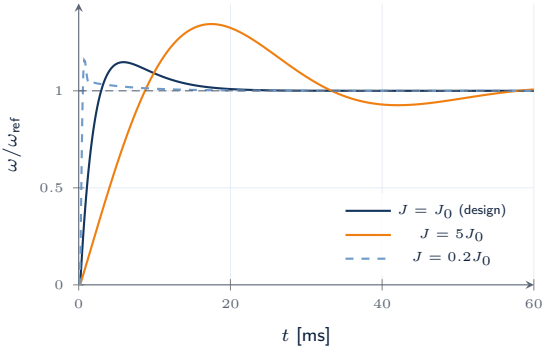
Session 3

2.8 Introduction to Adaptive and Robust Control

Why fixed gains are not enough: parameter variation

Parameter	Typical variation in servos
inertia J	robot arm pose, payload, winding roll: 1:3 . . . 1:10+
friction	temperature, lubrication, wear
torque const. k_t	magnet temperature, saturation at peak current
resonance f_r	axis position, load mass, belt tension
delay	sample rate, comms, filters

Velocity PI tuned for J_0 ($f_c = 100\,\text{Hz}$): $\omega_c \approx K_p/J$, so a $5\times$ inertia shifts f_c to $\approx 27\,\text{Hz}$ where the PI zero now costs $\approx 43^\circ \Rightarrow$ **34 % overshoot**.



Two answers: **adapt** the controller to the actual parameters, or design **one robust** controller that works for the whole set.

Describing model uncertainty

additive $G = G_0 + W_a \Delta$

multiplicative $G = G_0\,(1 + W\Delta)$

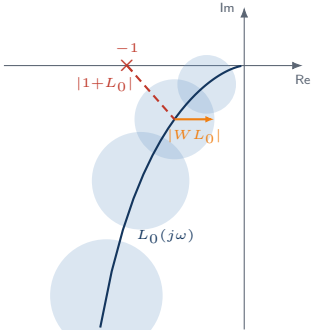
$\|\Delta\|_\infty \le 1$ (any stable Δ)

- **parametric**: $J \in [J_{\min}, J_{\max}]$ — structured
- **unstructured**: unmodelled resonances, delay, current loop — covered by a weight $|W(j\omega)|$ that **grows with frequency**

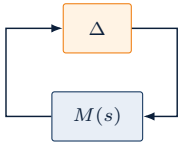
Robust stability (Nyquist): at each ω the true L lies in a disc of radius $|WL_0|$ around L_0. Stability for all members iff no disc covers -1:

$$|WL_0| < |1 + L_0| \iff |W(j\omega)\,T_0(j\omega)| < 1 \quad \forall\omega$$

[Skogestad & Postlethwaite, 2005, ch. 7]; [Zhou, Doyle & Glover, 1996].



Small-gain theorem, H_∞ idea and a robust-stability test



Small gain: M, Δ stable and

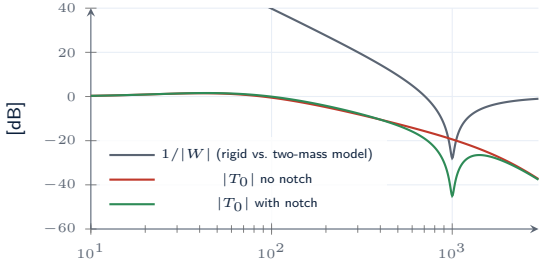
$$\|M\|_\infty \, \|\Delta\|_\infty < 1 \;\Rightarrow\; \text{loop stable}$$

$\|M\|_\infty = \sup_\omega |M(j\omega)|$ = peak gain. Multiplicative case: $M = WT_0 \Rightarrow |WT_0| < 1$.

H_∞ design: find C with

$$\left\| \begin{matrix} W_1 S \\ W_3 T \end{matrix} \right\|_\infty < 1$$

$S + T = 1$: not both small — performance at LF, robustness at HF.



$\max |WT_0| = 2.7$ without notch (test fails), 0.14 with notch (robustly stable).

[Zhou, Doyle & Glover, 1996]; [Skogestad & Postlethwaite, 2005]. Example: the 1 kHz resonance of Session 2.6.

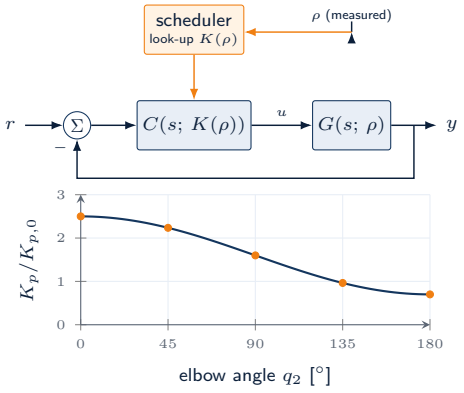
Gain scheduling

- design linear controllers at several operating points $\rho_1, \dots, \rho_m$
- measure a **scheduling variable** ρ: robot joint angle, payload, roll diameter, speed, DC-bus voltage
- interpolate gains between grid points

Servo example: keep ω_c constant while J changes:

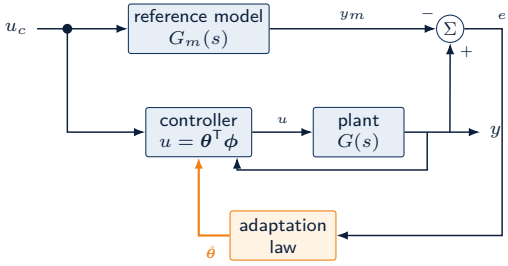
$$K_p(\rho) = \hat{J}(\rho)\,\omega_c, \qquad K_i(\rho) = K_p(\rho)\,\omega_c/4$$

- open-loop adaptation: no learning, fast, transparent, certifiable
- needs a good model of $J(\rho)$; ρ must vary **slowly** vs. loop dynamics
- switch gains bumplessly (rescale the integrator state)



illustrative inertia profile of a 2-link arm; dots = grid points

Model-reference adaptive control (MRAC): structure



- **reference model** G_m states the desired closed-loop behaviour
- **inner loop**: ordinary feedback with adjustable parameters $\hat{\boldsymbol{\theta}}$
- **outer loop**: slow adaptation drives $e = y - y_m \to 0$
- **direct** MRAC adapts controller gains; **indirect** (self-tuning) estimates J, B_v online and recomputes gains

Needs **persistent excitation** for parameters to converge; otherwise only the tracking error converges.

[Åström & Wittenmark, 1995, ch. 5]; [Ioannou & Sun, 1996].

The MIT rule (gradient adaptation)

Minimise $J(\theta) = \frac{1}{2}e^2$ by steepest descent:

$$\frac{d\theta}{dt} = -\gamma\, e\, \frac{\partial e}{\partial \theta}$$

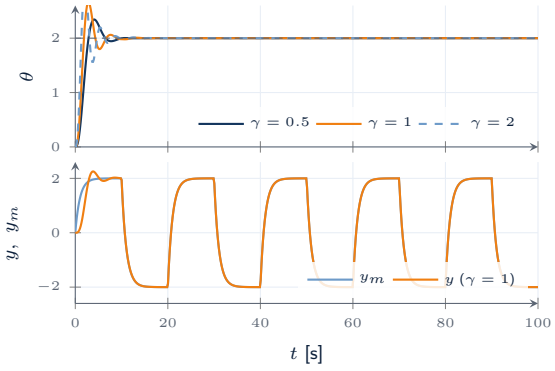
Adapting a feed-forward gain: plant $kG(s)$, model $k_0 G(s)$, $u = \theta u_c$, k unknown:

$$e = (k\theta - k_0)\, G u_c \;\Rightarrow\; \frac{\partial e}{\partial \theta} = kGu_c = \frac{k}{k_0} y_m$$

$$\frac{d\theta}{dt} = -\gamma'\, e\, y_m$$

Simulation: $G = \frac{1}{s+1}$, $k = 1$, $k_0 = 2 \Rightarrow \theta^* = 2$; $u_c =$ square wave.

- larger γ: faster, but **no stability guarantee** — large γu_c^2 can diverge



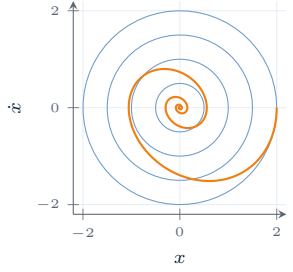
Classical example after [Åström & Wittenmark, 1995]; simulated for this course.

Lyapunov stability and Lyapunov-based adaptation

Lyapunov's idea: find an "energy" $V(\boldsymbol{x}) > 0$ with

$$\dot{V}(\boldsymbol{x}) < 0 \quad (\boldsymbol{x} \neq 0)$$

$\Rightarrow$ trajectories roll downhill: stable — without solving the ODE.



$\ddot{x} + 0.4\dot{x} + x = 0$, $V = x^2 + \dot{x}^2$, $\dot{V} = -0.8\dot{x}^2 \leq 0$

[Slotine & Li, 1991, ch. 3 & 8]; [Åström & Wittenmark, 1995].

First-order MRAC with stability proof (cf. velocity loop $J\dot{\omega} = -B_v\omega + u$): plant $\dot{y} = -ay + bu$, model $\dot{y}_m = -a_m y_m + b_m r$, control $u = \theta_1 r - \theta_2 y$. Ideal gains $\theta_1^* = b_m/b$, $\theta_2^* = (a_m - a)/b$; error $e = y - y_m$:

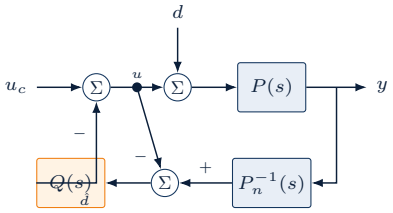
$$\dot{e} = -a_m e + b\,\tilde{\theta}_1 r - b\,\tilde{\theta}_2 y, \qquad \tilde{\theta}_i = \theta_i - \theta_i^*$$

$$V = \tfrac{1}{2}e^2 + \tfrac{b}{2\gamma}\left(\tilde{\theta}_1^2 + \tilde{\theta}_2^2\right)$$

$$\dot{\theta}_1 = -\gamma\, e\, r, \quad \dot{\theta}_2 = \gamma\, e\, y \;\Rightarrow\; \dot{V} = -a_m e^2 \leq 0$$

- stable for **any** $\gamma > 0$ (needs $\operatorname{sign} b$ known, $b > 0$ here)
- compare: MIT rule uses y_m, Lyapunov law uses the signals r, y — similar form, but proven
- Barbalat's lemma $\Rightarrow e \to 0$; $\tilde{\theta} \to 0$ only with persistent excitation

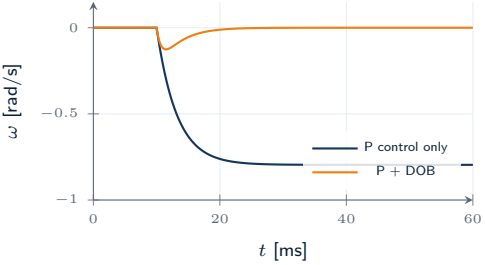
Disturbance observer (DOB)



$$\hat{d} = Q(s)\left[P_n^{-1}(s)\,y - u\right], \qquad u = u_c - \hat{d}$$

Velocity axis: $P_n^{-1} = J_n s$ (inverse inertia model), $Q = \frac{1}{\tau s+1}$ makes it proper.

[Ohnishi, Shibata & Murakami, 1996].



$J = 2 \times 10^{-3}$ kg m^2, P loop 50 Hz, Q: 200 Hz, load step 0.5 N m at 10 ms. Without DOB: -0.80 rad/s steady error; with DOB: $\to 0$.

- inside the Q bandwidth the plant behaves like $P_n \Rightarrow$ **robust to J, friction, T_L**
- Q bandwidth limited by noise, delay, resonances

Session 2.8 summary

Approach	Idea	Strength	Weakness
Robust (H_∞, small gain)	one fixed C for the whole uncertainty set: $\lvert WT\rvert < 1$	guaranteed, simple to run	conservative: less bandwidth
Gain scheduling	pre-computed gains vs. measured ρ	transparent, fast	needs model of $J(\rho)$, slow ρ
MRAC (MIT / Lyapunov)	adjust $\hat{\boldsymbol{\theta}}$ online so $y \to y_m$	copes with unknown parameters	excitation, transients, bursting
Disturbance observer	estimate lumped disturbance, cancel it	strong load rejection	Q bandwidth vs. noise/delay

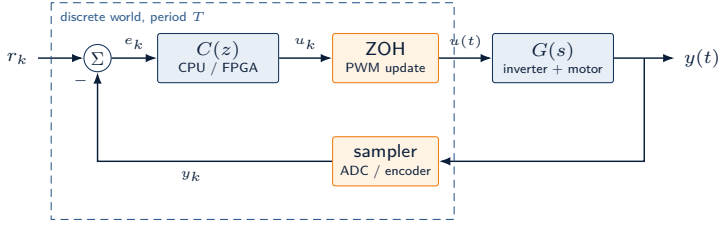
Key take-aways

- Model uncertainty grows with frequency: robust stability is decided around and above ω_c.
- Lyapunov functions prove stability without solving the dynamics — and give adaptation laws for free.
- In industrial drives: auto-tuning + gain scheduling + DOB/observers dominate; full MRAC is rare.

Session 4

2.9 Discrete-Time Control

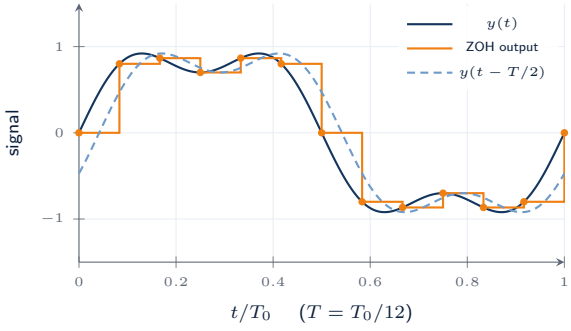
Anatomy of a digital servo loop



- signals exist only at $t = kT$; between samples the actuator **holds**
- quantisation: ADC bits, encoder counts, PWM resolution
- computation takes time: u_k is usually applied at $(k{+}1)T$

Loop	typical f_s	typical f_c
current	8–32 kHz	1–3 kHz
velocity	4–16 kHz	100–500 Hz
position	1–8 kHz	20–100 Hz

Sampling and the zero-order hold



[Franklin, Powell & Workman, 1998, ch. 4 & 6].

$$G_{\text{ZOH}}(s) = \frac{1 - e^{-sT}}{s}$$

$$G_{\text{ZOH}}(j\omega) = T\,\frac{\sin(\omega T/2)}{\omega T/2}\,e^{-j\omega T/2}$$

- the staircase follows $y(t - T/2)$ on average: **half-sample delay**
- phase lag $-\omega T/2$ (linear in ω)
- amplitude droop: $-0.9\,\text{dB}$ at $\omega = \omega_s/4$
- delay is invisible in the sampled values but *not* to the loop

Delay budget and phase loss at crossover

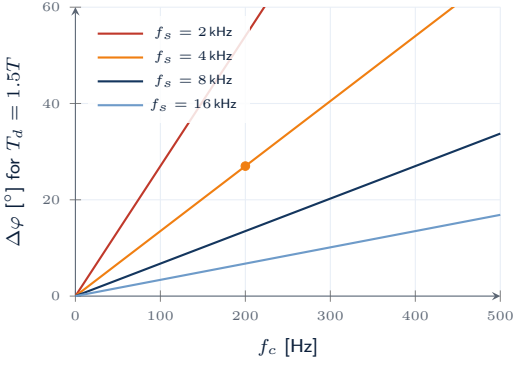
$$T_d \approx \underbrace{\frac{T}{2}}_{\text{ZOH}} + \underbrace{T_{\text{calc}}}_{\approx T} + T_{\text{filter}} + T_{\text{sensor}}$$

$$\Delta\varphi = \omega_c T_d \ [\text{rad}] = 360^\circ f_c T_d$$

Worked example. Velocity loop sampled at $f_s = 4\,\text{kHz}$, one-sample calculation delay: $T_d = 1.5T = 375\,\mu\text{s}$.

- $f_c = 200\,\text{Hz}$: $\Delta\varphi = 360 \cdot 200 \cdot 375 \cdot 10^{-6} = \mathbf{27^\circ}$
- at 8 kHz: 13.5°; at 16 kHz: 6.8°
- add a 1 kHz 1st-order LPF: $+\arctan(0.2) = 11.3^\circ$

Delay is the dominant phase thief in digital drives — double f_s, halve the loss.

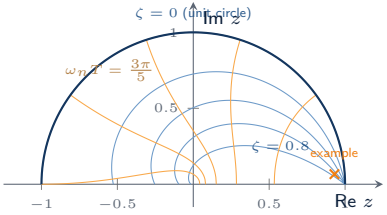


From the s-plane to the z-plane

$$z = e^{sT} \qquad \text{stable} \iff |z_i| < 1$$

s-plane	z-plane
left half-plane	inside unit circle
$j\omega$ axis	unit circle
$s = 0$	$z = 1$
$\omega = \pi/T$ (Nyquist)	$z = -1$

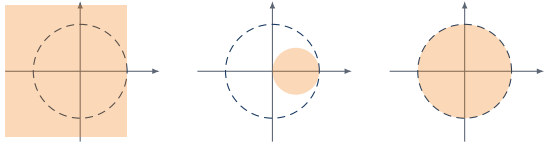
- closer to the origin $\Rightarrow$ faster decay
- poles near $z = -1$: ringing at $f_s/2$
- **example:** $s = -70 \pm j71.4$, $T = 1\,\text{ms}$ $\Rightarrow z = e^{-0.07}e^{\pm j0.0714} = 0.930 \pm j0.0665$ — fast sampling crowds poles near $z = 1$ (numerics!)



blue-grey: constant $\zeta = 0, 0.2, \ldots, 0.8$; amber: constant $\omega_n T = k\pi/5$

Discretisation methods

Method	Substitution / rule	Properties	Use
Forward Euler	$s \leftarrow \dfrac{z-1}{T}$	stable s may map unstable ($\omega_f T > 2$)	avoid for fast poles
Backward Euler	$s \leftarrow \dfrac{z-1}{zT}$	always stable; distorts, adds damping	integrators, D-term
Tustin (bilinear)	$s \leftarrow \dfrac{2}{T}\dfrac{z-1}{z+1}$	LHP $\leftrightarrow$ unit disc exactly; frequency warping	PI/PID, filters (pre-warp)
ZOH equivalent	$G(z) = (1-z^{-1})\,\mathcal{Z}\{G(s)/s\}$	exact at samples for staircase input	plant models
Pole–zero matching	$z_i = e^{s_i T}$, match DC gain	preserves pole locations	filters, notches



orange: image of the stable s half-plane; dashed: unit circle

Example: LPF pole $\omega_f = 2\pi \cdot 500$, $T = 250\,\mu s$ ($\omega_f T = 0.785$):

exact / ZOH / matched	$z_p = e^{-\omega_f T} = 0.456$
Tustin	0.436
backward Euler	0.560
forward Euler	0.215

At $f_f = 1.5$ kHz, forward Euler gives $z_p = -1.36$: **unstable**.

Tustin frequency warping — why notch filters need pre-warping

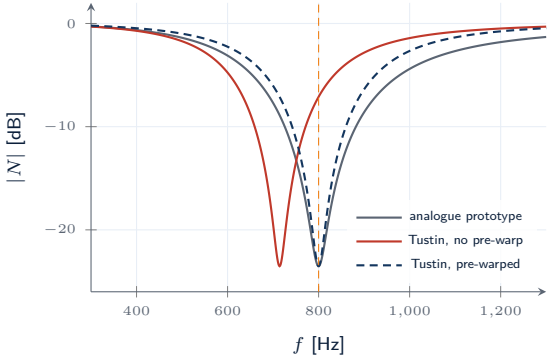
Tustin maps the analogue frequency ω_a to the digital frequency ω_d by

$$\omega_a = \frac{2}{T}\tan\frac{\omega_d T}{2}$$

Fine for $\omega \ll \omega_s$, but compresses everything towards $f_s/2$.

Example: 800 Hz notch, $f_s = 4$ kHz: without pre-warp the notch lands at $\frac{2}{T}\arctan\frac{\omega_0 T}{2} = 2\pi\cdot$**714 Hz** ($-11\,\%$) and misses the resonance.

Pre-warp: design the analogue prototype at $\omega_{pw} = \frac{2}{T}\tan\frac{\omega_0 T}{2} \Rightarrow$ exact at 800 Hz.



[Oppenheim & Schafer, 2010, ch. 7]; [Franklin, Powell & Workman, 1998, ch. 6].

Discrete PID implementation

$C(s) = K_p + \dfrac{K_i}{s} + \dfrac{K_d s}{T_f s + 1}$, period T:

$$I_k = I_{k-1} + K_i T\, e_k \qquad \text{(backward Euler)}$$
$$D_k = \tfrac{T_f}{T_f + T} D_{k-1} - \tfrac{K_d}{T_f + T} (y_k - y_{k-1})$$
$$u_k = \mathrm{sat}(K_p e_k + I_k + D_k)$$

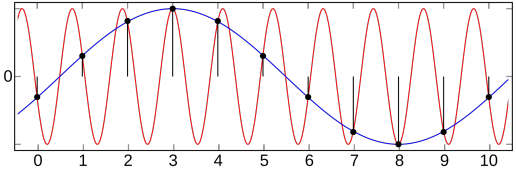
- D acts on $-y$ (no derivative kick on reference steps)
- **anti-windup**: stop or back-calculate I when u saturates
- **incremental form** Δu_k: bumpless switching
- fixed-point: scale $K_i T$ carefully (tiny numbers)

[Åström & Hägglund, 2006, ch. 3]; [Franklin, Powell & Workman, 1998].

```
// executed every T (ISR synchronised to PWM)
e  = r - y;
I += Ki*T*e;                              // integrate
D  = a*D - b*(y - y_old);                 // a=Tf/(Tf+T)
y_old = y;                                // b=Kd/(Tf+T)
u  = Kp*e + I + D;
if (u > UMAX)      { I -= u - UMAX; u = UMAX; }
else if (u < -UMAX){ I -= u + UMAX; u = -UMAX;}
write_pwm(u);       // takes effect next period
```

Anti-windup above: clamping by back-calculation with gain 1 (integrator absorbs the excess).

Aliasing: a teaser



Moxfyre, CC BY-SA 3.0, via Wikimedia Commons

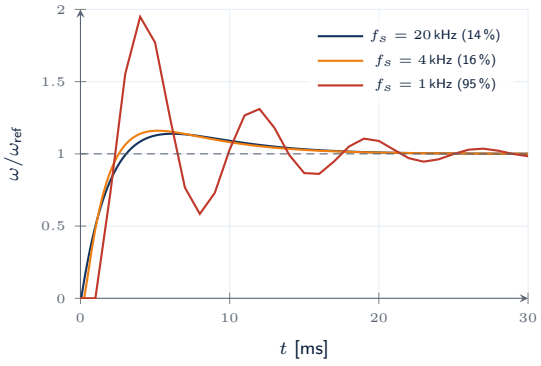
red: $f = 0.9$, blue: $f = 0.1$, sampled at $f_s = 1$ — identical samples

$$f_{\text{alias}} = |f - k\,f_s|\,, \quad k = \text{round}(f/f_s)$$

- anything above $f_s/2$ folds back into $[0, f_s/2]$
- **example:** 4.6 kHz structural/switching component sampled at 4 kHz $\Rightarrow$ appears at **600 Hz** — inside the loop's range
- after sampling it cannot be separated: filter *before* the sampler
- remedies: analogue RC anti-alias filter, oversampling + decimation, sampling synchronous to the PWM centre

More in Day 5 (sensing and signal acquisition).

Choosing the sample rate



[Franklin, Powell & Workman, 1998, ch. 11]; simulation for this course.

Same velocity PI ($f_c = 100\,\text{Hz}$, $J = 2 \times 10^{-3}\,\text{kg m}^2$), discrete, 1-sample calculation delay.

At 1 kHz: $T_d = 1.5\,\text{ms} \Rightarrow \Delta\varphi = 54^\circ \Rightarrow$ near-unstable.

Rules of thumb

- $f_s \geq 20$–$40\times$ closed-loop bandwidth for near-continuous behaviour
- check $360^\circ f_c T_d$ fits the PM budget
- nested loops: inner loops sample faster
- sensor and PWM rate must match f_s

Session 2.9 summary

Key take-aways

- Sampling + ZOH + calculation $\approx 1.5T$ delay $\Rightarrow$ phase loss $360^\circ f_c T_d$.
- Stability in z: all poles inside the unit circle; $z = e^{sT}$.
- Tustin for controllers (pre-warp notches), ZOH-equivalent for plants, avoid forward Euler for fast poles.
- Discrete PID: backward-Euler I, filtered D on measurement, anti-windup, saturation.
- Filter analogue signals before sampling (aliasing).
- Choose f_s from the bandwidth *and* the phase budget.

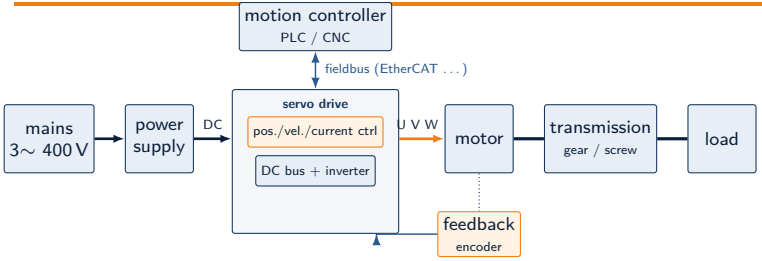
f_s (with $f_c = 200$ Hz)	$\Delta\varphi$ at $T_d = 1.5T$
2 kHz	54°
4 kHz	27°
8 kHz	13.5°
16 kHz	6.8°

Next: how these loops are stacked inside a real servo drive.

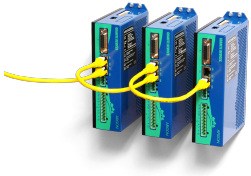
Session 5

3.1 Servo System Architecture

Servo system architecture: from mains to load



- **drive** = power stage + control loops + protection; **feedback** closes every loop
- the motion controller sends set-points (position/velocity/torque) every 0.25–4 ms



Tero Kontkanen, CC BY-SA 3.0, via Wikimedia Commons

compact servo drives on a daisy-chained fieldbus;



Projuktiponno, CC BY-SA 4.0, via Wikimedia Commons

incremental encoder

Torque, velocity and position control modes

Mode	Command from controller	Loops closed in the drive	Typ. BW	Applications
Torque (current)	T^* or i_q^*	current loop only	1–3 kHz	winders/tension, press-fit force, master–slave torque sharing
Velocity	ω^*	velocity + current	100–500 Hz	spindles, conveyors, CNC with position loop in the controller
Position	θ^* (or profile)	position + velocity + current	20–100 Hz	point-to-point, robots, synchronised axes

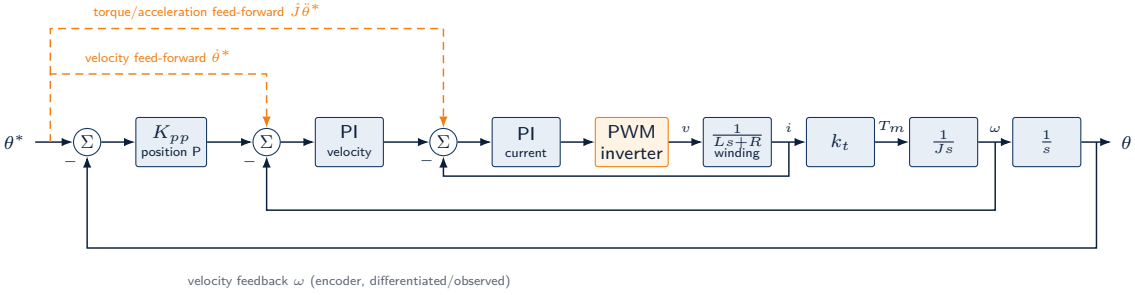
Where is each loop closed?

- cyclic synchronous modes (CiA 402 / IEC 61800-7): **CSP**, **CSV**, **CST** — set-points every cycle
- profile modes (PP, PV): the drive generates the trajectory itself
- the faster the inner loop, the closer it must sit to the power stage

Selection rule

Close a loop where its **sensor** and its **update rate** are; send the controller only what it can update fast enough.

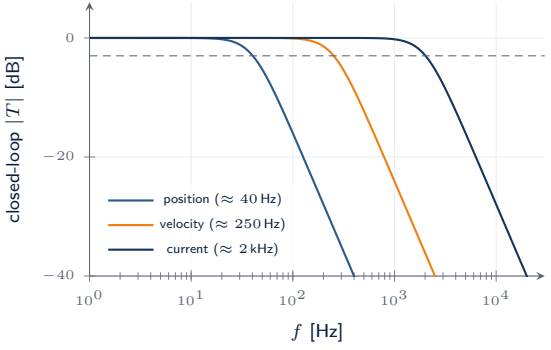
Cascade control structure of a servo drive



- each inner loop makes the outer plant simple: current loop → "torque source", velocity loop → "integrator"
- bandwidth separation **3–10×** between loops
- feed-forwards give tracking without high feedback gains
- tuned inside-out: current → velocity → position

[Ellis, 2012, ch. 17]; [Leonhard, 2001].

Session 3.1 summary: the bandwidth hierarchy



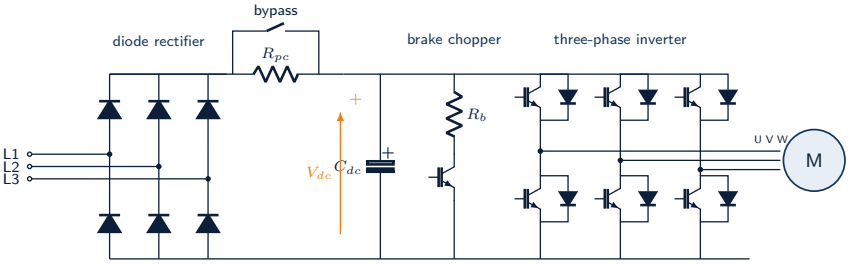
Key take-aways

- Servo = supply + drive + motor + transmission + load + feedback + motion controller.
- Three modes: torque, velocity, position — defined by which loops the drive closes.
- Cascade: current $\gg$ velocity $\gg$ position bandwidth (factor 3–10 each).
- Feed-forward for tracking, feedback for disturbances.

Session 6

3.2 Power Electronics I: Rectifier, DC Bus and Inverter

Power stage of a servo drive: overview



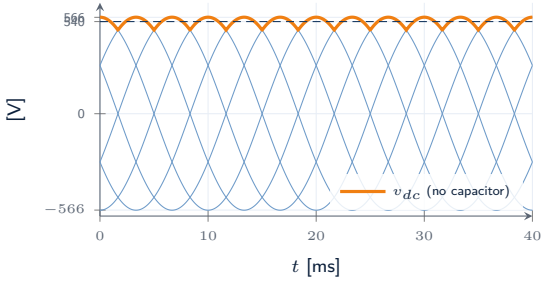


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Terminals: L1/L2(/L3) mains, L1C/L2C control supply, B1/B2 brake resistor, $\oplus$/$\ominus$ DC bus, U/V/W motor

AC $\rightarrow$ DC $\rightarrow$ AC: uncontrolled rectifier charges the DC link; pre-charge resistor limits inrush; the brake chopper dumps regenerated energy; the two-level inverter synthesises variable-voltage, variable-frequency three-phase voltages by PWM.

Rectifier and DC link



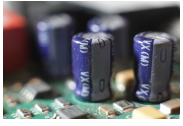
6-pulse output = envelope of the six line-to-line voltages; ripple at $6f = 300$ Hz.

$$V_{dc,\text{avg}} = \frac{3\sqrt{2}}{\pi} V_{LL} = 1.35\, V_{LL}, \qquad \hat{V} = \sqrt{2}\, V_{LL}$$

400 V mains: 540 V average, 566 V peak (light load, capacitor charges to peak).

- C_{dc}: ripple current, energy buffer, low-inductance commutation loop
- pre-charge resistor + bypass relay: limit inrush into an empty C_{dc}
- diode front end is **not regenerative**





left: Hiro36, CC BY-SA 3.0; right: Giovanna 27, CC BY 4.0; via Wikimedia Commons

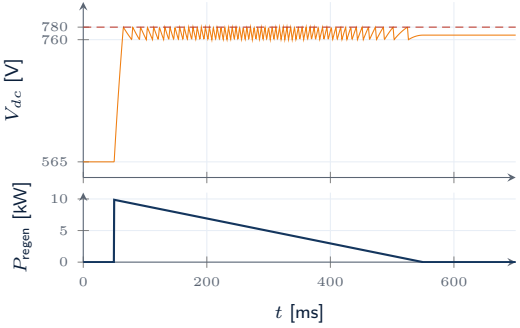
Regeneration and the brake chopper

Worked example. $J = 0.05\,\text{kg m}^2$ decelerated linearly from 3000 rpm (314 rad/s) to 0 in 0.5 s:

- kinetic energy $E = \frac{1}{2} J\omega^2 = 2467$ J
- torque $J\alpha = 31.4$ N m, peak power $P = T\omega = 9.87$ kW
- $C_{dc} = 1$ mF from 565 to 780 V absorbs only $\frac{1}{2}C(780^2 - 565^2) = 145$ J (6 %)
- resistor must take peak power at V_{on}: $R_b \leq V_{on}^2/P = 61.6\,\Omega \Rightarrow$ choose **56 Ω**

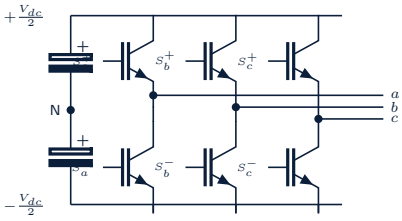


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Hysteresis chopper: on at 780 V, off at 760 V. Alternatives: common DC bus between axes, active (regenerative) front end.

Three-phase two-level inverter



antiparallel (freewheeling) diodes omitted for clarity — see photo

- each leg ties its phase to $+V_{dc}/2$ or $-V_{dc}/2$: **two-level**
- S^+ and S^- of a leg switch **complementarily** — never both on
- $2^3 = 8$ switching states: 6 active, 2 zero vectors
- average leg voltage set by duty: $\bar{v}_{aN} = (2d_a - 1)V_{dc}/2$
- freewheeling diodes carry current during dead time and regeneration



Sthdhk, CC BY-SA 4.0, via Wikimedia Commons

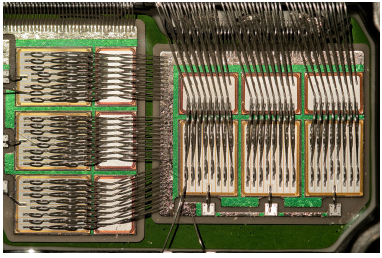
dual (half-bridge) IGBT module

Power switches: IGBT, MOSFET and wide band-gap

	Si IGBT	Si MOSFET	SiC MOSFET
voltage class	600–1700 V	$\leq$ 250 V (low-voltage drives)	650–1700 V
conduction	$V_{CE(sat)} \approx$ 1.5–2 V	$R_{DS(on)}$: mΩ	$R_{DS(on)}$, low
switching	tail current, slower	very fast	very fast
typ. f_{sw}	4–16 kHz	20–100 kHz	16–100 kHz
servo use	400 V industrial drives	24–72 V axes, robots, AGVs	compact/high-f_s drives

- higher $f_{sw} \Rightarrow$ less current ripple, faster current loop, smaller filters
- but high dv/dt stresses motor insulation and causes bearing currents and EMI

[Mohan, Undeland & Robbins, 2003, ch. 8 & 22]; [Krishnan, 2001].



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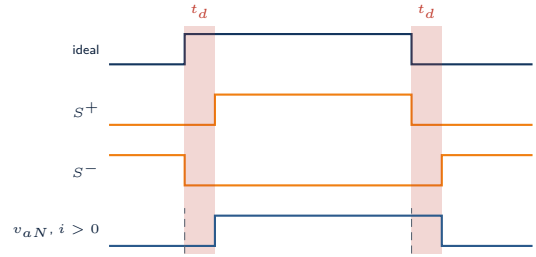
opened IGBT module: paralleled IGBT dies and freewheeling diodes with bond wires

Gate drivers and dead time (preview)

Gate driver functions

- level shift / **galvanic isolation** for the high-side switch (bootstrap or isolated supply)
- peak gate current of several A; typical IGBT drive $+15\,\mathrm{V}$ / $-8\,\mathrm{V}$
- **desaturation** (short-circuit) detection, soft turn-off within $\approx 10\,\mu\mathrm{s}$
- Miller clamp, under-voltage lock-out

Dead time t_d: both switches off after each transition to prevent **shoot-through**.



During t_d the diode conducts, so v_{aN} depends on the *current sign*: for $i > 0$ the pulse is shortened by t_d.

$$\Delta\bar{v} \approx \frac{t_d}{T_{sw}}\, V_{dc}\ \mathrm{sign}(i)$$

Example: $t_d = 2\,\mu\mathrm{s}$, $f_{sw} = 10\,\mathrm{kHz}$, $V_{dc} = 560\,\mathrm{V} \Rightarrow \Delta\bar{v} = \mathbf{11.2\,V}$ — distortion at low speed; compensation on Day 4.

[Balogh, TI SLUA618]; [Mohan, Undeland & Robbins, 2003].

Session 3.2 (part 1) summary

Key take-aways

- Drive power path: rectifier → pre-charge → DC link → brake chopper → inverter → motor.
- $V_{dc} \approx 1.35 V_{LL}$ (540 V at 400 V mains).
- Braking energy mostly exceeds what C_{dc} can store: size R_b from $V_{on}^2 / P_{\text{peak}}$.
- Two-level inverter: 8 states, complementary switching per leg.
- Gate driver: isolation, protection; dead time prevents shoot-through but costs volts.

Quantity	Value (400 V drive)
V_{dc} nominal	540–565 V
chopper on	≈780 V
over-voltage trip	≈800–850 V
IGBT class	1200 V
f_{sw}	4–16 kHz
dead time	0.5–3 µs

typical orders of magnitude; check the specific drive datasheet

Day 3 summary

Control

- in-loop filters cost phase; notches tame resonances; lead adds PM, lag adds stiffness
- state feedback + observer: pole placement, separation, integral state, LQR
- robust: $|WT| < 1$; adaptive: MRAC with MIT or Lyapunov laws; scheduling and DOB in practice
- digital: $T_d \approx 1.5T$, $z = e^{sT}$, Tustin + pre-warp, anti-windup

Servo hardware

- servo = drive + motor + transmission + feedback + motion controller
- torque / velocity / position modes, cascaded loops with feed-forward
- rectifier, DC link, brake chopper, two-level inverter, gate drivers, dead time

One sentence

The achievable servo bandwidth is set by phase: filters, delay and resonances spend it, compensators and fast sampling buy it back.

Preview of Day 4

Day 4 — Servo drives in depth

- PWM techniques: sinusoidal PWM, space-vector modulation, over-modulation
- dead-time effects and compensation (continuing today's preview)
- field-oriented control: Clarke/Park transforms, i_d/i_q current loops, decoupling
- current, velocity and position loop tuning in practice
- feed-forward, resonance handling and auto-tuning in commercial drives

Bring with you: the delay-budget formula $360° f_c T_d$, the notch-filter parameters and the cascade structure — Day 4 tunes exactly these loops on a PMSM drive.

Figure credits

All other diagrams and plots: own drawings, computed for this course (TikZ / pgfplots / circuitikz).

File	Source (Wikimedia Commons)	Author, licence
d3_aliasing.png	File:AliasingSines.svg	Moxfyre, CC BY-SA 3.0
d3_servo_drives.png	File:Servo motor drive.png	Tero Kontkanen, CC BY-SA 3.0
d3_encoder.jpg	File:Rotary Encoder.jpg	Projuktiponno, CC BY-SA 4.0
d3_servo_drive_motor.jpg	File:Yeskawa Servo Drive & Motor.jpg	Projuktiponno, CC BY-SA 4.0
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d3_brake_resistor.jpg	File:Turbrake.jpg	Glogger at English Wikipedia, CC BY-SA 3.0
d3_igbt_module.jpg	File:MITSUBISHI IGBT module.jpg	Sthdhk, CC BY-SA 4.0
d3_igbt_module_open.jpg	File:CM600DU-24NFH.jpg	Pslawinski, CC BY-SA 3.0

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